



Finite Languages and Automata June Exam 2019

Formal Languages and Automata (University of the Witwatersrand, Johannesburg)



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Course or topic No(s)

COMS3003

Course or topic name(s)
Paper Number & title

Formal Languages and Automata

Exam to be
held during the months(s) of

13 June 2019

Year of Study

Degrees/Diplomas for which
this course is prescribed

BSc, BEconSc, BCom

Faculty/ies presenting
candidates

Science; Commerce, Law and Management

Internal examiner(s)
and telephone extension
number(s)

Prof. S. Ewert x76180

External examiner(s)

Dr Christopher Cleghorn, UP

Special materials required

None

Time allowance

Course nos	COMS3003	Hours	3
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Instructions to candidates

There are 80 marks available. 80 marks = 100%. You have 2,25 minutes per mark.

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Examination

Formal Languages and Automata

student number:

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June 2019

Instructions

- Answer all the questions on the question paper.
- Answer each question in the space allocated for the answer immediately following the question. If you run out of space, you may continue on the last page; in that case, remember to give the number of the question.

Comments

Let $x \in \{a, b\}^*$ be an arbitrary string. Then,

- $n_a(x)$ indicates the total number of occurrences of the symbol a in the string x .
- x^r indicates the reverse of the string x .
- $\tilde{x}$ denotes the string obtained by replacing all a 's by b 's and vice versa. For example, $\tilde{\lambda} = \lambda$ and $\tilde{abb} = baa$.

For lecturer's use only

Question no.	Maximum marks	Marks obtained
1	3	
2	3	
3	3	
4	20	
5	9	
6	8	
7	19	
8	15	

Questions: please turn over

Question 1 — Languages in general

Let $\mathcal{L}_1 = \{a\}$ and $\mathcal{L}_2 = \{aa\}$.

In each of the following you are given a set and two possible answers, labelled **empty** and **non empty**, respectively. Circle either **empty** or **non empty**; moreover, if you choose the answer labelled **non empty**, complete the sentence.

a. $(\mathcal{L}_1^* \cap \mathcal{L}_2^*) \setminus (\mathcal{L}_1 \cap \mathcal{L}_2)^*$

(i) **empty**: The set is empty.

(ii) **non empty**: The set contains the string _____.

b. $(\mathcal{L}_1 \cap \mathcal{L}_2)^* \setminus (\mathcal{L}_1^* \cap \mathcal{L}_2^*)$

(i) **empty**: The set is empty.

(ii) **non empty**: The set contains the string _____.

[Total marks for Question 1 = 3 marks]

Question 2 — Regular expressions

Consider the following regular expressions.

- a. $(0 + 10)^* 1^*$
- b. $(1 + 0)^* (11(1 + 0)^* 010 + 010(1 + 0)^* 11)(1 + 0)^*$
- c. $(\lambda + 0)(1 + 10)^*$
- d. $(00 + 11 + (01 + 10)(00 + 11)^*(01 + 10))^*$
- e. $(1 + 01)^*(\lambda + 0)$
- f. $(\lambda + 0 + 00)(1 + 10 + 100)^*$
- g. $(1 + 0)^*(0^+1^+0^+ + 1^+0^+1^+)(1 + 0)^*$

Consider the following languages over $\{0, 1\}^*$. In each case, give the label of a regular expression in the list above that corresponds to the language.

- a. The language of all strings that do not contain the substring 110. _____
- b. The language of all strings in which both the number of 0's and 1's are even. _____
- c. The language of all strings containing both 01 and 10 as substrings. _____

[Total marks for Question 2 = 3 marks]

Question 3 — Regular and Nonregular Languages

- a. Let $\mathcal{L} \subseteq \Sigma^*$ be any language. Let $[x]$ denote the equivalence class of $I_{\mathcal{L}}$ containing x . We want to show that if $[x]$ contains a string xy , where $y \neq \lambda$, then it also contains xyy . Complete the following proof.

$xI_{\mathcal{L}}xy$. Then for all strings $z \in \Sigma^*$ of the form

$z =$ _____ ,

either both $xz \in \mathcal{L}$ and $xyz \in \mathcal{L}$, or both $xz \notin \mathcal{L}$ and $xyz \notin \mathcal{L}$. Therefore $xyI_{\mathcal{L}}xyy$.

1 mark(s)

- b. In each of the following cases, give two different strings that are indistinguishable with respect to $\mathcal{L}$, if such strings exist. If there are no such strings, state it clearly.

(i) $\mathcal{L} = \{0^n y \mid n \geq 0, y \in \{0, 1\}^* \text{ and } |y| \leq n\}$.

1 mark(s)

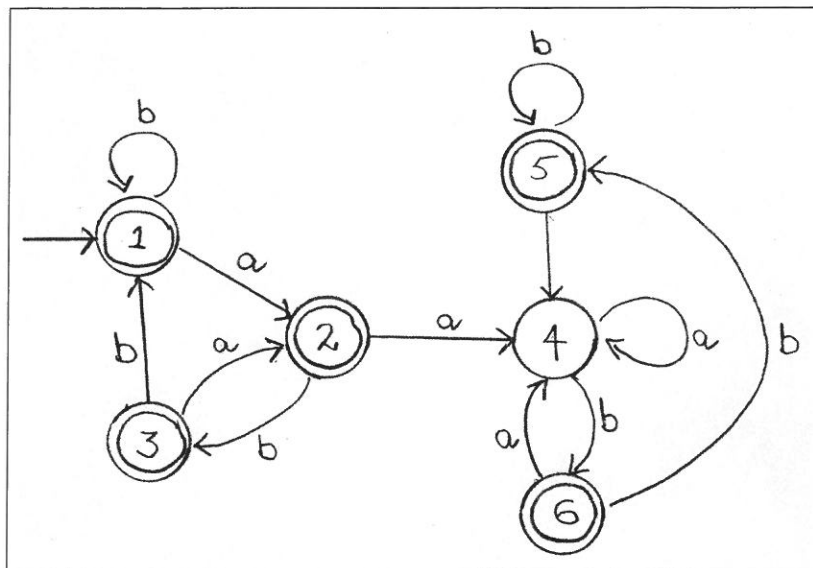
(ii) $\mathcal{L} = \{x \mid x = x^r\}$.

1 mark(s)

[Total marks for Question 3 = 3 marks]

Question 4 — Finite Automata and Decision Algorithms

a. Consider the following FA M that accepts some language $\mathcal{L}$:



In the process of finding the minimum-state FA $M_{\min}$ that accepts $\mathcal{L}$, we obtain the following diagram after completion of the final pass of the algorithm for marking the pairs of states (p, q) with $p \neq q$. Here X indicates that a pair is marked and an empty block indicates that a pair is not marked.

2	X				
3		X			
4	X	X	X		
5	X	X	X	X	
6	X	X	X	X	
q	1	2	3	4	5

On the following page, give $M_{\min}$ as a diagram. In each state of $M_{\min}$, indicate clearly which states of M it represents.

8 mark(s)

Please answer Question 4.a here.

c. Answer the following question:

For languages $\mathcal{L}_1$ and $\mathcal{L}_2$, it is given that $\mathcal{L}_1 \cup \mathcal{L}_2$ is regular and that $\mathcal{L}_1$ is finite. Can we conclude from this that $\mathcal{L}_2$ is regular?

Use the following

$$\mathcal{L}_2 = ((\mathcal{L}_1 \cup \mathcal{L}_2) \cap \mathcal{L}'_1) \cup (\mathcal{L}_1 \cap \mathcal{L}_2) ,$$

where $\mathcal{L}'_1$ indicates the complement of $\mathcal{L}_1$.

6 mark(s)

[Total marks for Question 4 = 20 marks]

Question 5 — The Pumping Lemma for Regular Languages

a. Consider the language

$$\mathcal{L} = \{xx^r \mid x \in \{a,b\}^+\}.$$

We want to prove that $\mathcal{L}$ is not regular.

Suppose $\mathcal{L}$ is regular. According to the pumping lemma for regular languages, there exists an n such that for every $x \in \mathcal{L}$ with $|x| \geq n$, x can be written uvw for some u, v and w with $|uv| \leq n$, $|v| > 0$, and for any $m \geq 0$, $uv^mw \in \mathcal{L}$.

Let n be the integer in the statement of the pumping lemma.

Choose x so that $|x| \geq n$.

Complete the proof.

$x =$ _____.

Then $uv =$ _____.

This implies that $w =$ _____.

Moreover, $v =$ _____.

According to the lemma, $uv^mw \in \mathcal{L}$ for every $m \geq 0$.

Consider

$$\begin{aligned} uv^mw &= \dots \\ &= \dots \\ &= \dots \end{aligned}$$

For $m =$ _____, $uv^mw \notin \mathcal{L}$,

since _____

This contradicts the lemma, therefore our assumption that $\mathcal{L}$ is regular is wrong.

8 mark(s)

- b. Which lemma (named after a scientist) uses markings on the letters in a word and is more powerful than the lemma above?

1 mark(s)

[Total marks for Question 5 = 9 marks]

Question 6 — Context-Free Languages

a. Consider the following sets of production rules.

- (i) $\{S \rightarrow aSc \mid T, T \rightarrow aTb \mid \lambda\}$
- (ii) $\{S \rightarrow AB, A \rightarrow aAb \mid \lambda, B \rightarrow bBc \mid \lambda\}$
- (iii) $\{S \rightarrow AT \mid UC, A \rightarrow aA \mid \lambda, C \rightarrow cC \mid \lambda, T \rightarrow bTc \mid \lambda, U \rightarrow aUb \mid \lambda\}$
- (iv) $\{S \rightarrow TC \mid U, C \rightarrow cC \mid \lambda, T \rightarrow aTb \mid \lambda, U \rightarrow aUc \mid B, B \rightarrow bB \mid \lambda\}$
- (v) $\{S \rightarrow TBC \mid AU, T \rightarrow aTb \mid \lambda, B \rightarrow bB \mid b, C \rightarrow cC \mid \lambda\} \cup \{A \rightarrow aA \mid a, U \rightarrow aUc \mid V, V \rightarrow bV \mid \lambda\}$
- (vi) $\{S \rightarrow aaSb \mid aSb \mid Sb \mid \lambda\}$
- (vii) $\{S \rightarrow aS_1b \mid S_1b, S_1 \rightarrow aaS_1b \mid aS_1b \mid S_1b \mid \lambda\}$
- (viii) $\{S \rightarrow aSb \mid aSbb \mid \lambda\}$

Consider the following languages over $\{a, b, c\}$. In each case, give the label of a set of production rules in the list above that would generate the language, assuming the start symbol to be S .

- (i) $\{a^i b^j c^k \mid j = i + k\}$

1 mark(s)

- (ii) $\{a^i b^j c^k \mid i = j \text{ or } j = k\}$

1 mark(s)

- (iii) $\{a^i b^j \mid j \leq 2i\}$

1 mark(s)

Question 7 — Pushdown Automata

- a. Consider the language $\mathcal{L}$ of all palindromes over $\{a, b\}$. Consider the Pushdown Automaton M that accepts $\mathcal{L}$, where

$$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, Z_0\}, q_0, Z_0, \{q_2\}, \delta) ,$$

and δ is given in the following table:

Move #	State	Input	Stack symbol	Move(s)	
				(i)	(ii)
1	q_0	a	Z_0	(q_0, aZ_0)	(q_1, Z_0)
2	q_0	b	Z_0	(q_0, bZ_0)	(q_1, Z_0)
3	q_0	a	a	(q_0, aa)	(q_1, a)
4	q_0	b	a	(q_0, ba)	(q_1, a)
5	q_0	a	b	(q_0, ab)	(q_1, b)
6	q_0	b	b	(q_0, bb)	(q_1, b)
7	q_0	λ	Z_0	(q_1, Z_0)	
8	q_0	λ	a	(q_1, a)	
9	q_0	λ	b	(q_1, b)	
10	q_1	a	a	(q_1, λ)	
11	q_1	b	b	(q_1, λ)	
12	q_1	λ	Z_0	(q_2, Z_0)	

Answer the following questions:

- (i) Give the number of one move in δ that doesn't use any input symbol.

1 mark(s)

- (ii) Give the number of one move in δ that reads an input symbol, but doesn't store it on the stack.

1 mark(s)

- (iii) Give the number of one move in δ that erases an a on top of the stack.

1 mark(s)

- (iv) Give the number of one move in δ that reads an input symbol b and stores it on the stack.

1 mark(s)

(v) Consider the word $w = ababa$.

1. Give one sequence of moves on w that leads to w being accepted. Remember to give the number of every move.

6 mark(s)

2. Give one sequence of moves on w that leads to w not being accepted. Remember to give the number of every move.

2 mark(s)

b. Consider the language

$$\mathcal{L} = \{a^i b^j \mid i \leq j \leq 2i; i, j \geq 0\}.$$

We want to construct a PDA

$$M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, b, Z_0\}, q_0, Z_0, \{q_0, q_3\}, \delta)$$

that accepts $\mathcal{L}$.

Complete the following transition table.

7 mark(s)

Move no.	State	Input	Stack symbol	Move(s)
1	q_0	a	Z_0	
2	q_1	a	a	
3	q_1	b	a	
4	q_2	b	a	
5	q_2	λ	Z_0	
all other combinations				none

[Total marks for Question 7 = 19 marks]

Question 8 — The Pumping Lemma for Context-free Languages

- a. The pumping lemma for context-free languages (CFLs) states:

Let $\mathcal{L}$ be a CFL. Then there is an integer n such that for any $u \in \mathcal{L}$ with $|u| \geq n$, there are strings v, w, x, y , and z satisfying $u = vwx y z$, $|w y| > 0$, $|w x y| \leq n$, and for any $m \geq 0$, $vw^m x y^m z \in \mathcal{L}$.

Using the lemma, derive the following statement:

Suppose $\mathcal{L}$ is an infinite context-free language. Then there are integers p and q , with $q > 0$, such that for every $m \geq 0$, $\mathcal{L}$ contains a string of length $p + mq$.

2 mark(s)

- b. Let n and u be as defined in the pumping lemma for context-free languages above. Consider the following languages. In each case, give an appropriate value for u .

(i) $\mathcal{L} = \{s \in \{a, b, c\}^+ \mid n_a(s) \geq n_b(s) \geq n_c(s) \geq 1\}$

1 mark(s)

(ii) $\mathcal{L} = \{s \in \{a, b, c\}^+ \mid n_a(s) > n_b(s) > n_c(s) \geq 1\}$

1 mark(s)

(iii) $\mathcal{L} = \{s \in \{a, b, c\}^+ \mid 1 \leq n_a(s) = n_b(s), n_c(s) > n_a(s) + n_b(s)\}$

1 mark(s)

c. Consider

$$\mathcal{L} = \{s \in \{a, b, c\}^* \mid n_a(s) \geq n_b(s) + n_c(s), n_b(s) = n_c(s)\}.$$

We want to prove that $\mathcal{L}$ is not context-free.

Suppose $\mathcal{L}$ is context-free. According to the pumping lemma, there is an integer n so that for any $u \in \mathcal{L}$ with $|u| \geq n$, there are strings v, w, x, y , and z satisfying $u = vwx y z$, $|w y| > 0$, $|w x y| \leq n$, and for any $m \geq 0$, $vw^m x y^m z \in \mathcal{L}$.

Let n be the integer in the statement of the pumping lemma.

Choose $u \in \mathcal{L}$ so that $|u| \geq n$; we choose

$$u = a^{2n} b^n c^n.$$

Suppose v, w, x, y , and z are any strings satisfying the conditions above.

Complete the following parts of the proof.

If w and y contain

(i) a 's only, then $u' = vw^m x y^m z \notin \mathcal{L}$ for

$m =$ _____ ,

since _____

3 mark(s)

(ii) b 's and c 's, but no a 's, then $u' = vw^m x y^m z \notin \mathcal{L}$ for

$m =$ _____ ,

since _____

3 mark(s)

(iii) c 's only, then $u' = vw^m x y^m z \notin \mathcal{L}$ for

$m =$ _____ ,

since _____

3 mark(s)

- d. For the language $\mathcal{L}$ in Question 8.c above, give one bad choice for u , in other words, one value for u such that $u \in \mathcal{L}$ and $|u| \geq n$, but one will not be able to get a contradiction for all possible values for w and y .

1 mark(s)

[Total marks for Question 8 = 15 marks]

End of question paper